MIL-C-92B
24 February 1969
SUPERSEDING
MIL-C-92A
2 April 1959

MILITARY SPECIFICATION

CAPACITORS, VARIABLE, AIR DIELECTRIC (TRIMMER), GENERAL SPECIFICATION FOR

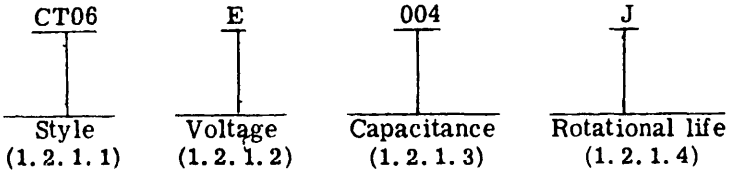
This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for trimmer, air-dielectric, variable capacitors intended primarily for use in high-frequency trimming applications. These capacitors are suitable for operation over a temperature range of -55° to +85° C.

1.2 Classification.

1.2.1 Part number. The part number shall be in the following form, and as specified (see 3.1 and 6.1):



1.2.1.1 Style. The style is identified by the two-letter symbol "CT"; the letters identify trimmer, air-dielectric, variable capacitors, and the number identifies the shaft type and shaft length or dimensions or both of the capacitors (see 6.3).

1.2.1.2 Voltage. The direct-current (dc) rated voltage is identified by a single letter in accordance with table I.

TABLE I. DC rated voltages.

Symbol	Voltage
	Volts
A	50
B	100
C	300
D	350
E	500
F	600
G	700

1.2.1.3 Capacitance. The nominal maximum capacitance value expressed in picofarads (pf) is identified by a three-digit number. For values of 1 to 9 pf, inclusive, the first two digits shall be zeros, and for values of 10 uuf to 99 pf inclusive, the first digit shall be a zero.

1.2.1.4 Rotational life. The rotational life is identified by a single letter in accordance with table II.

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TABLE II. Rotational life.

Symbol	Rotational life
	Cycles
J	250
M	10,000

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-G-23827 - Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-C-39028 - Capacitors, Packaging of.
MIL-C-45662 - Calibration System Requirements.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.

See supplement 1 for list of applicable specification sheets.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Specification sheets. The individual part requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Capacitors furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for the opening of the bids (see 4.4 and 6.2).

3.3 Materials. When a definite material is not specified, a material shall be used which will enable the capacitors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

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3.3.1 Plating. When silver plating is used, it shall be provided with a corrosion resistant film such as a chromate conversion coating.

3.4 Design and construction. Capacitors shall be of the design, construction, and physical dimensions specified (see 3.1).

3.4.1 Threaded parts. All threaded parts shall be class 2A and 2B fit after plating, in accordance with Handbook H28.

3.4.2 Terminals. The terminals shall be permanently secured. All external joints shall be soldered or welded. All terminals used for soldered connections shall be hot-tin-dipped or solder-coated per soldering requirements of MIL-STD-454 to facilitate soldering (see 3.22.1).

3.4.3 Shaft. The shafts shall be made of metal. The rotor shall be free to rotate through 360°

3.4.4 Bushing of lock-type capacitors. In order to prevent rotation of the capacitor bushing within the ceramic base, the bushing shall be secured in one of two ways:

(a) By fitting a multisided shoulder or collar into the multisided recess of the ceramic base if the shoulder or collar is an integral part of the bushing.
(b) By spinning or staking the end of the bushing over a multisided, metal washer and fitting the bushing into the multisided recess of the ceramic base. The spun or staked end of the bushing shall be soldered to the washer. When this is inconsistent with the fabrication of the capacitor, the end of the bushing shall be spun or staked in such a manner that the formed end shall follow the contour of the recess to such an extent that the sides of the recess shall prevent the bushing from turning.

3.5 Dielectric withstanding voltage. Capacitors shall withstand the potential specified in 4.6.2 without damage, arcing, or breakdown.

3.6 Barometric pressure. Capacitors shall withstand the potential specified in 4.6.3 without damage, arcing, or breakdown.

3.7 Insulation resistance. When measured as specified in 4.6.4, the insulation resistance shall be not less than 100,000 megohms.

3.8 Capacitance. When measured as specified in 4.6.5, the minimum capacitance shall be not more and the maximum not less than the values specified. The capacitance range may be broader than that specified provided it encompasses the range specified (see 3.1).

3.9 Quality factor (Q). When determined as specified in 4.6.6, the Q shall exceed the value specified (see 3.1).

3.10 Capacitance change versus rotation. When measured as specified in 4.6.7, the capacitance change versus rotation of the rotor shall not deviate from a straight line by more than 10 percent, and shall show no reversals in direction.

3.11 Contact resistance. When measured as specified in 4.6.8, the contact resistance shall not exceed 0.01 ohm.

3.12 Torque. When measured as specified in 4.6.9, the torque required to start and turn the the rotor shall be within the limits specified (see 3.1). As a result of the test specified in 4.6.9, there shall be no visible mechanical damage.

3.13 Capacitance (at high and low temperature). When measured as specified in 4.6.10, the capacitance change shall not exceed 0.5 percent of the initial value or 0.4 pf, whichever is greater.

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3.14 Low-temperature exposure. When capacitors are tested as specified in 4.6.11, there shall be no evidence of fractures, loosening of parts, or other visible mechanical damage.

3.15 Temperature cycling. When capacitors are tested as specified in 4.6.12, there shall be no evidence of fractures, loosening of parts, or other visible mechanical damage.

3.16 Salt spray (corrosion). When tested as specified in 4.6.13, capacitors shall meet the following requirements:

Dielectric withstanding voltage	As specified in 3.5.
Insulation resistance	Not less than 3,000 megohms.
Capacitance	As specified in 3.8.
Q	As specified in 3.9.
Contact resistance	Shall not exceed 0.01 ohm.
Torque (at 25° ±5°C only)	As specified in 3.12.

There shall be no evidence of corrosion of exposed metallic surfaces in excess of 10 percent. Discoloration and tarnishing alone are not sufficient cause for rejection.

3.17 Moisture resistance. When tested as specified in 4.6.14, capacitors shall meet the following requirements:

Dielectric withstanding voltage	As specified in 3.5.
Insulation resistance	Not less than 50,000 megohms.
Capacitance	As specified in 3.8.
Contact resistance	As specified in 3.11.
Torque (at 25° ±5°C only)	As specified in 3.12.

There shall be no evidence of corrosion of exposed metallic surfaces in excess of 10 percent.

3.18 Shock, medium impact. When capacitors are tested as specified in 4.6.15, there shall be no momentary or intermittent contacts of 0.5 millisecond (ms) or greater duration, open or short-circuiting or any evidence of fractures, loosening of parts, or other visible mechanical damage. Capacitance change shall be not more than 0.2 percent or 0.2 pf, whichever is greater, from initial value.

3.19 Vibration, high frequency. When capacitors are tested as specified in 4.6.16, there shall be no momentary or intermittent contacts of 0.5 ms or greater duration, open or short-circuiting, or any evidence of fracture, loosening of parts, or other visible mechanical damage. The capacitance change shall be not more than 0.2 percent or 0.2 pf, whichever is greater, from the initial value obtained when measured as specified in 4.6.5.

3.20 Rotational life. When tested as specified in 4.6.17, capacitors shall meet the following requirements:

Dielectric withstanding voltage	As specified in 3.5.
Contact resistance	As specified in 3.11.
Torque (at 25° ±5°C only)	As specified in 3.12.

3.21 Marking. Capacitors shall be permanently and legibly marked with the part number and the manufacturer's name or symbol. Marking shall not be placed on the mounting surface. Paper labels shall not be used. Other markings which in any way interfere with, obscure, or confuse those specified herein are prohibited. The marking shall remain legible after all tests.

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3.21.1 Part number. There shall be no space between the symbols which comprise the part number. When the size of the capacitor does not permit the part number to be marked on one line, it may be divided into two lines as shown in the following example:

CT06
E004J

3.22 Workmanship. Capacitors shall be processed in such a manner as to be uniform in quality, and shall meet the requirements of 3.3 to 3.4.4 inclusive. All rough edges shall be eliminated from all metallic parts and the unit shall be free from other defects that will affect life, serviceability, or appearance.

3.22.1 Soldering. Rosin, or rosin and alcohol, shall be used to solder connections. Corrosive fluxes may be used only to solder the plates on the rotor or stator supports. After completion of soldering operation, corrosive fluxes shall be completely neutralized.

3.22.2 Lubrication. Lubricating grease shall conform to MIL-G-23827.

3.23 Resistance to soldering heat (soldered terminals only). When capacitors are tested as specified in 4.6.18, there shall be no damage to the capacitor which will cause any electrical or mechanical failure of the capacitor. Capacitance change shall not exceed 0.2 percent or 0.2 pf, whichever is greater, from initial value; Q shall be as specified in 3.9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. The supplier shall establish and maintain a calibration system in accordance with MIL-C-45662.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

(a) Qualification inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Inspection conditions and methods. Unless otherwise specified herein, all inspection shall be made at a temperature of 25° ±5°C, a relative humidity between 20 and 80 percent, and barometric pressure of 28 to 32 inches of mercury.

4.3.1 Capacitor setting. Unless otherwise specified herein, the rotor plates of all capacitors shall be set at the maximum capacitance setting and maintained at this position throughout the tests.

4.4 Qualification inspection.

4.4.1 Sample size. The number of specimens comprising a sample of capacitors to be submitted for qualification inspection shall be as specified in the appendix to this specification.

4.4.2 Inspection routine. The specimens shall be subjected to the examinations and tests specified in table III, in the order shown. All specimens shall be subjected to the examinations and tests in group I. The specimens shall then be divided into the remaining groups shown in table III, and subjected to the tests for their particular group.

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4.4.3 Failures. Failures in excess of those allowed in table III shall be cause for refusal to grant qualification.

TABLE III. Qualification inspection.

Examination or test	Requirement paragraph	Method paragraph	Number of specimens to be inspected	Number of failures allowed 1/
Group I				
Visual and mechanical examination 2/	3.1, 3.3 to 3.4.4 incl, and 3.21 to 3.22.2 incl	4.6.1	37 3/	1
Dielectric withstanding voltage	3.5	4.6.2		
Barometric pressure	3.6	4.6.3		
Insulation resistance	3.7	4.6.4		
Capacitance	3.8	4.6.5		
Quality factor (Q)	3.9	4.6.6		
Capacitance change versus rotation	3.10	4.6.7		
Contact resistance	3.11	4.6.8		
Torque	3.12	4.6.9		
Group II				
Capacitance (at high and low temperature)	3.13	4.6.10	12	1
Low-temperature exposure	3.14	4.6.11		
Temperature cycling	3.15	4.6.12		
Salt spray (corrosion)	3.16	4.6.13		
Group III				
Moisture resistance	3.17	4.6.14	12	1
Group V				
Resistance to soldering heat (soldered terminals only)	3.23	4.6.18	12	1
Shock, medium impact	3.18	4.6.15		
Vibration, high frequency	3.19	4.6.16		
Rotational life	3.20	4.6.17		

1/ A specimen having one or more defects shall be considered as a single failure.

2/ Marking shall be considered as a defect only if it becomes illegible as a result of any one of the tests.

3/ One additional specimen is included in each sample of 37 specimens to permit substitution for the allowable failure in group I.

4.4.4 Retention of qualification. To retain qualification, the supplier shall forward, at 6-month intervals, to the qualifying activity, a summary of the results of groups A and B tests, indicating as a minimum the number of lots which passed and the number which failed, and a summary of the results of group C tests, including the number and type of any part failures. The summary shall include those tests performed during that 6-month period. If the summary of the test results indicates nonconformance with specification requirements, action shall be taken to remove the failing product from the qualified products list. Failure to submit the summary shall result in loss of qualification for that product. In addition to the periodic submission of inspection data, the supplier shall immediately notify the qualifying activity at any time during the 6-month period that the inspection data indicates failure of the qualified product to meet the requirements of the specifica

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4.5 Quality conformance inspection. Quality conformance inspection shall consist of groups A, B, and C inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspection. Except as specified in 4.5.2.1.4, delivery of products which have passed the groups A and B inspection shall not be delayed pending the results of group C inspection.

4.5.1.1 Inspection lot. An inspection lot shall be as specified in MIL-STD-105, and shall consist of capacitors of the same style, voltage, and rotational life.

4.5.1.2 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table IV, in the order shown.

4.5.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality levels (AQL) shall be as specified in table IV. Major and minor defects shall be as defined in MIL-STD-105.

4.5.1.2.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

TABLE IV. Group A inspection.

Examination or test	Requirement paragraph	Method paragraph	AQL (percent defective)	
			Major	Minor
Visual and mechanical examination:				
Material	3.3	4.6.1	1.0	4.0
Body dimensions	3.4	---		
Design and construction (other than body dimensions)	3.4 to 3.4.4 incl	---		
Marking 1/	3.21	---		
Workmanship	3.22 to 3.22.2 incl	---		
Dielectric withstanding voltage	3.5	4.6.2	1.0	---
Barometric pressure	3.6	4.6.3		
Insulation resistance	3.7	4.6.4		
Capacitance	3.8	4.6.5		
Quality factor (Q)	3.9	4.6.6		
Capacitance change versus rotation	3.10	4.6.7		
Contact resistance	3.11	4.6.8		
Torque (at 25° ±5°C only)	3.12	4.6.9		

1/ Marking defects shall be based on visual examination only.

4.5.1.3 Group B inspection. Group B inspection shall consist of the tests specified in table V, in the order shown and shall be performed on sample units which have been subjected to and have passed the Group A inspection. The group B sample size shall be based on the same lot size that group A was based on.

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TABLE V. Group B inspection.

Test	Requirement paragraph	Method paragraph
Resistance to soldering heat (soldered terminals only)	3.23	4.6.18
Torque	3.12	4.6.9
Rotational life	3.20	4.6.17

4.5.1.3.1 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105 for small sample inspection using the special inspection levels, unless otherwise specified herein. Normal inspection shall be used at the start of the contract. The AQL shall be 4.0 (percent defective) and the inspection level S-4.

4.5.1.3.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.1.3.3 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on the contract or order.

4.5.2 Periodic inspection.

4.5.2.1 Group C inspection. Group C inspection shall consist of the tests specified in table VI, in the order shown. A separate sample shall be used for each sampling period and for each subgroup listed.

TABLE VI. Group C inspection.

Test	Requirement paragraph	Method paragraph
Every 2 weeks		
Capacitance (at high and low temperature)	3.13	4.6.10
Low-temperature exposure	3.14	4.6.11
Salt spray (corrosion)	3.16	4.6.13
Every month		
Subgroup 1		
Moisture resistance	3.17	4.6.14
Subgroup 2		
Shock, medium impact	3.18	4.6.15
Vibration, high frequency	3.19	4.6.16

4.5.2.1.1 Sampling plan. Sample units shall be tested every 2 weeks and every month in accordance with table VI. The number of sample units to be selected for each sampling period and subgroup shall be in accordance with table VII. Sample units shall be selected at random throughout each sampling period as to be representative of the product produced during this interval. Different types may be combined to form the total number of units produced, and a proportionate number of sample units selected from each type.

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4.5.2.1.2 Defectives. If the number of defectives exceed the number allowed in table VII, the sample shall be considered to have failed.

TABLE VII. Selection of sample units and defectives allowed.

Number of units produced during the sampling period	Sample size	Number of defectives allowed 1/
Up to 100 inclusive	2	0
101 to 800 inclusive	5	0
801 to 3,200 inclusive	15	1
3,201 to 8,000 inclusive	25	2
8,001 to 22,000 inclusive	35	3
22,001 and over	50	4

1/ A sample unit having one or more defects shall be considered a single defective.

4.5.2.1.3 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract or order.

4.5.2.1.4 Noncompliance. If a sample fails to pass group C inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government). Groups A and B inspection may be reinstituted; however, final acceptance shall be withheld until the group C reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the contracting officer.

4.5.3 Inspection of preparation for delivery. Sample packages or packs and the inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of MIL-C-39028.

4.6 Methods of examination and test.

4.6.1 Visual and mechanical examination. Capacitors shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.3 to 3.4.4, inclusive, and 3.21 to 3.22.2, inclusive).

4.6.2 Dielectric withstanding voltage (see 3.5). Capacitors shall be tested in accordance with method 301 of MIL-STD-202. The following details shall apply:

(a) Magnitude and nature of test voltage - 200 percent of the specified dc rated voltage or an alternating-current (ac) (rms) voltage equal to 142 percent of the specified dc working voltage (see table I and 3.1).
(b) Points of application of test voltage - The test voltage shall be applied between the rotor and stator terminals; and between the rotor and stator terminals electrically connected together and the mounting hardware.
(c) Limiting value of surge current - Surge current shall not exceed 50 milliamperes.
(d) Measurements after dielectric withstanding voltage test - After the test, capacitors shall be visually examined for evidence of damage, arcing, or breakdown.

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4.6.3 Barometric pressure (see 3.6). Capacitors shall be tested in accordance with method 105 of MIL-STD-202. The following details shall apply:

(a) Method of mounting - Optional. There shall be no side pressure applied to displace the shaft from its normal position during the tests.
(b) Test condition - As specified (see 3.1).
(c) Tests during subjection to reduced pressure - The specified voltage (see table I and 3.1), shall be applied to the capacitors between the rotor and stator terminals, and between the rotor and stator terminals electrically connected together and the mounting hardware, for a period of 1 minute.
(d) Measurements - After the test, capacitors shall be visually examined for evidence of damage, arcing, or breakdown.

4.6.4 Insulation resistance (see 3.7). Capacitors shall be tested in accordance with method 302 of MIL-STD-202. The following details shall apply:

(a) Test potential - A potential equal to the dc rated voltage or 500 volts, whichever is less, shall be applied.
(b) Points of measurement - Measurements shall be made between the rotor and stator terminals; and between the rotor and stator terminals electrically connected together and the mounting hardware.

4.6.5 Capacitance (see 3.8). The minimum and maximum capacitance shall be measured in accordance with method 305 of MIL-STD-202. The following detail shall apply:

Test frequency - 1 ± 0.1 megahertz (MHz).

4.6.6 Quality factor (Q) (see 3.9). Capacitors shall be tested in accordance with method 306 of MIL-STD-202. The following detail shall apply:

Test frequency - Capacitors shall be set at minimum capacitance, and the Q shall be measured at a frequency of 1 ± 0.1 MHz.

4.6.7 Capacitance change versus rotation (see 3.10). Capacitors shall be set at approximately 10 percent of the maximum capacitance value above the minimum capacitance value, and the rotor shall then be rotated in steps of 30° until 90 percent of maximum capacitance is reached. Capacitance shall be measured after each step at a frequency of 1 ± 0.1 MHz. Accuracy of the test equipment shall be within ± 1 percent.

4.6.8 Contact resistance (see 3.11). Capacitors shall be tested in accordance with method 307 of MIL-STD-202. The following details and exception shall apply:

(a) Method of connection - Contact resistance shall be measured between the rotor terminal and the rotor shaft.
(b) Test current - 0.5 ampere dc.
(c) Number of activations prior to measurement - Not applicable.
(d) Number of test activations - One.
(e) Number of measurements per activation - Three. Measurements shall be made at approximately minimum, midrange, and maximum capacitance settings.

4.6.9 Torque (see 3.12). Unless otherwise specified, torque measurements shall be made at $85° \begin{smallmatrix}+3° \\ -0°\end{smallmatrix}$, $25° \pm 5°$, and $-55° \begin{smallmatrix}+0° \\ -3°\end{smallmatrix}$C. After the test, capacitors shall be visually examined for evidence of mechanical damage.

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4.6.9.1 Fixed-torque type capacitor. A gradually increasing torque shall be applied to the rotor. The torque required to start and maintain rotation shall be measured.

4.6.9.2 Lock-type capacitors. The torque of lock-type capacitors shall be measured as specified in 4.6.9.1 with the locknut loose. The locknut shall then be tightened by a force of 100 ounce-inches. The locknut shall be loosened and tightened six times and, while locked, the torque required to turn the rotor shall be measured.

4.6.10 Capacitance (at high and low temperature) (see 3.13). Capacitance measurements shall be made as specified in 4.6.5, except that measurements shall be made at -55° ±3° and +85° ±3°C. The measurement at each temperature shall be recorded when two successive readings taken at 5-minute intervals indicate no change in capacitance.

4.6.11 Low-temperature exposure (see 3.14). Capacitors shall be exposed to a temperature of -62° ±5° C for a periof of 24 hours. After exposure, capacitors shall be visually examined for evidence of fractures, loosening of parts, or other visible mechanical damage.

4.6.12 Temperature cycling (see 3.15). Capacitors shall be tested in accordance with method 102 of MIL-STD-202. The following details shall apply:

(a) Test condition - D.
(b) Measurements after cycling - Capacitors shall be visually examined for evidence of fractures, loosening of parts, or other visible mechanical damage.

4.6.13 Salt spray (corrosion) (see 3.16). Capacitors shall be tested in accordance with method 101 of MIL-STD-202. The following details shall apply:

(a) Test condition - B.
(b) Measurements after exposure - Dielectric withstanding voltage, insulation resistance, capacitance, Q, contact resistance, and torque (at 25° ±5°C only), shall be measured as specified in 4.6.2, 4.6.4, 4.6.5, 4.6.6, 4.6.8, and 4.6.9, respectively. Capacitors shall also be visually examined for evidence of corrosion.

4.6.14 Moisture resistance (see 3.17). Capacitors shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

(a) Initial measurements - Not applicable.
(b) Mounting - Capacitors shall be mounted on a metal panel by their normal mounting means.
(c) Loading voltage - Not applicable.
(d) Final measurements - After 24 hours drying at ambient conditions, dielectric withstanding voltage, insulation resistance, capacitance, contact resistance, and torque (at 25° ±5°C only) shall be measured as specified in 4.6.2, 4.6.4, 4.6.5, 4.6.8, and 4.6.9, respectively. Capacitors shall also be visually examined for evidence of corrosion.

4.6.15 Shock, medium impact (see 3.18). Capacitors shall be tested in accordance with method 205 of MIL-STD-202. The following details and exceptions shall apply:

(a) Mounting - Capacitors shall be mounted by their normal mounting means.
(b) Test condition - C.

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has been touted as the ultimate bonded medium for ... Fiber optics. The use of FO

(c) Electrical loading during shock - During the test, a potential of 125 percent of the specified dc rated voltage (see table I and 3.1), shall be applied between the rotor and stator terminals on one half of the specimens, and between the rotor and stator terminals connected together and the mounting hardware 1/ on the other half of the specimens.

(d) Measurements during and after shock - During the test, observations shall be made for momentary or intermittent arcing. After the test and prior to dismounting the capacitor, capacitance shall again be measured as specified in 4.6.5. Capacitors shall be visually examined for evidence of breakdown, arcing, fractures, loosening of parts, or other visible mechanical damage. A cathode-ray oscilloscope or other comparable means shall be used as an indicating device in determining electrical damage.

4.6.16 Vibration, high frequency (see 3.19). Capacitors shall be tested in accordance with method 204 of MIL-STD-202. The following details and exceptions shall apply:

(a) Mounting and preparation of specimens - Capacitors shall be rigidly mounted on a vibration jig. The jig shall be constructed so as to preclude any resonance within the test range. An examination of the jig shall be made on a vibrator. If any resonant frequencies are observed, adequate steps must be taken to damp the structure. Lock-type capacitors shall be locked during the test.

(b) Measurements prior to vibration - Capacitors shall be set at approximately 90 percent of the fully meshed position, and capacitance shall be measured as specified in 4.6.5.

(c) Electrical load conditions - During the test, while the capacitors are set as specified in (b) above, a potential of 125 percent of the specified dc rated voltage (see table I and 3.1), shall be applied between the rotor and stator terminals on one-half of the specimens, and between the rotor and stator terminals connected together and the mounting hardware 1/ on the other half of the specimens.

(d) Test condition - B, with the following exceptions:
1. Duration of vibration - 12 hours.
2. Direction of motion - Capacitors shall be vibrated on each of three mutually perpendicular directions (4 hours in each direction).

(e) Measurements during and after vibration - During the test, observations shall be made for momentary or intermittent arcing. After the test and prior to dismounting the capacitor, capacitance shall again be measured as specified in 4.6.5; the capacitors shall also be visually examined for evidence of breakdown, arcing, fractures, loosening of parts, or other visible mechanical damage. A cathode-ray oscilloscope or other comparable means shall be used as an indicating device in determining electrical damage.

4.6.17 Rotational life. Capacitors shall be mounted in such a manner that the shafts are continously rotated backward and forward through at least 95 percent of the capacitance range for the applicable number of cycles (see table II and 3.1), at a rate of 25 ± 5 cycles per minute. After cycling, dielectric withstanding voltage, contact resistance, and torque (at 25° ±5°C only) shall be measured as specified in 4.6.2, 4.6.8, and 4.6.9, respectively (see 3.20).

4.6.18 Resistance to soldering heat (soldered terminals only) (see 3.23). Capacitors shall be tested in accordance with method 210 of MIL-STD-202. The following details shall apply:

(a) Test condition - B.
(b) Depth of immersion in the molten solder - 0.125 inch.

1/ A potential applied between the mounting hardware and the rotor and stator terminals connected together does not apply to style CT16 units.

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(c) Cooling time prior to final examinations and measurements - 10 minutes.
(d) Examination and measurements before and after test - Capacitors shall be set at approximately 90 percent. Capacitors shall be examined for evidence of mechanical or electrical damage. Capacitance and Q shall be measured as specified in 4.6.5 and 4.6.6, respectively.

5. PREPARATION FOR DELIVERY

5.1 Preparation. Capacitors shall be prepared for delivery in accordance with MIL-C-39028.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable specification sheet, and the complete part number (see 1.2.1 and 3.1).
(c) Levels of preservation and packaging and packing, and applicable marking (see section 5).

6.1.1 Indirect shipments. The preservation, packaging, packing, and marking requirements specified in section 5 apply only to direct purchases by or direct shipment to the Government and are not intended to apply to contracts or orders between the supplier and prime contractor.

6.2 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable qualified products list whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is U.S. Army Electronics Command; however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center, Directorate of Engineering Standardization (DESC-E), Dayton, Ohio 45401.

6.2.1 Copies of "Provisions Governing Qualification" may be obtained upon application to U.S. Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.3 Supersession data. This specification supersedes MIL-C-92A, dated 2 April 1959, and JAN-C-92, dated 30 December 1944. The characteristic, as specified in the type designation of JAN-C-92, is not included in MIL-C-92B. The information pertaining to characteristics B, C, E, and H is now covered by style number as shown in table VIII. Dash numbers of MIL-C-92/6 have been deleted and now are identified as shown in table IX.

TABLE VIII. Replacements for characteristics.

Characteristic of JAN-C-92	Replaced by style in MIL-C-92B
B - - - - - - - -	CT12
C - - - - - - - -	CT13
E - - - - - - - -	CT14
H - - - - - - - -	CT15

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TABLE IX. Replacement of former dash numbers.

MIL-C-92/6 dash number	Part number to be used in MIL-C-92/6A
M92/6-01	CT16D005J
M92/6-02	CT16D011J
M92/6-03	CT16D016J

6.4 Style CT16 units have a temperature coefficient of plus 45 ± 15 parts per million per degree centigrade.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL
Navy - EC
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - None
Navy - AS, OS, CG
Air Force - 19

Preparing activity:
Army - EL

(Project 5910-0833)

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APPENDIX

PROCEDURE FOR QUALIFICATION INSPECTION

10. SCOPE

10.1 This appendix details the procedure for submission of samples, with related data, for qualification inspection of capacitors covered by this specification. The procedure for extending qualification of the required samples to other capacitors covered by this specification is also outlined herein.

20. SUBMISSION

20.1 Sample. A sample consisting of 37 specimens of the type for which qualification is sought shall be submitted (see 30.1).

20.2 Test data. Each submission shall be accompanied by test data covering the nondestructive tests listed in table III which have been performed on the submitted specimens. The performance of the destructive tests by the supplier on a duplicate set of specimens is encouraged, although not required. All test data shall be submitted in duplicate.

20.3 Certification of material. When submitting samples for qualification, the supplier shall submit certification, in duplicate, that the materials used in his components are in accordance with the applicable specification requirements.

20.4 Description of items. The supplier shall submit a detailed description of the capacitors being submitted for inspection, including the design and construction, and the materials for the base, rotor, and electrodes.

30. EXTENT OF QUALIFICATION

30.1 Qualification will be restricted to the styles and rotational life specified in table X, and the voltage rating and capacitance values equal to and less than those of the type submitted.

TABLE X. Extent of qualification.

Style submitted	Style qualified
CT14 - - - - - -	CT14 and CT15
CT15 - - - - - -	CT14 and CT15
CT16 - - - - - -	CT16
Rotational life submitted	Rotational life qualified
J - - - - - - - -	J
M - - - - - - - -	M, J

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-92B Capacitors, Variable, Air Dielectric(Trimmer), General Specification For

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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